



Cloud Computing: Concept, Challenges and Trends

Prof. Dr. Rodrigo da Rosa Righi
rrrighi@unisinos.br

Applied Computing Graduate Program
Universidade do Vale do Rio dos Sinos - Brazil

Antes da Cloud

TCO considerations

1	Server Costs	Hardware: Server, rack chassis power distribution units (PDUs), top-of-rack (TOR) switches (and maintenance)	Software: Operating system (OS), virtualization licenses (and maintenance)	Facilities cost		
				Space	Power	Cooling
2	Storage Costs	Hardware: Storage disks, storage area network (SAN) or Fibre Channel (FC) switches	Storage administration costs	Facilities cost		
				Space	Power	Cooling
3	Network Costs	Network hardware: Local area network (LAN) switches, load balancer bandwidth costs	Network administration costs	Facilities cost		
				Space	Power	Cooling
4	IT Labor Costs	Server administration costs				

Outline

- * Today in Some Numbers
- * Cloud Computing at a Glance
- * Challenges in Cloud Computing
- * Research Projects
- * Trends in Cloud Adoption

Research and
Development
(R&D)

30 billion pieces of content were added to Facebook this past month.

Worldwide IP traffic will quadruple by 2025.

More than 2 billion videos were watched on YouTube ... yesterday.

The average teenager sends 4,762 text messages per month.

32 billion searches were performed last month ... on Twitter.

Observed Features

Storage

Performance

Scalability

Applications

Transparent Access



Traditional Infrastructure



Equipment



Resources and
administration



Contracts



Cost



AWS Cloud



No upfront
expense—pay for
what you use



Improve time to
market and agility



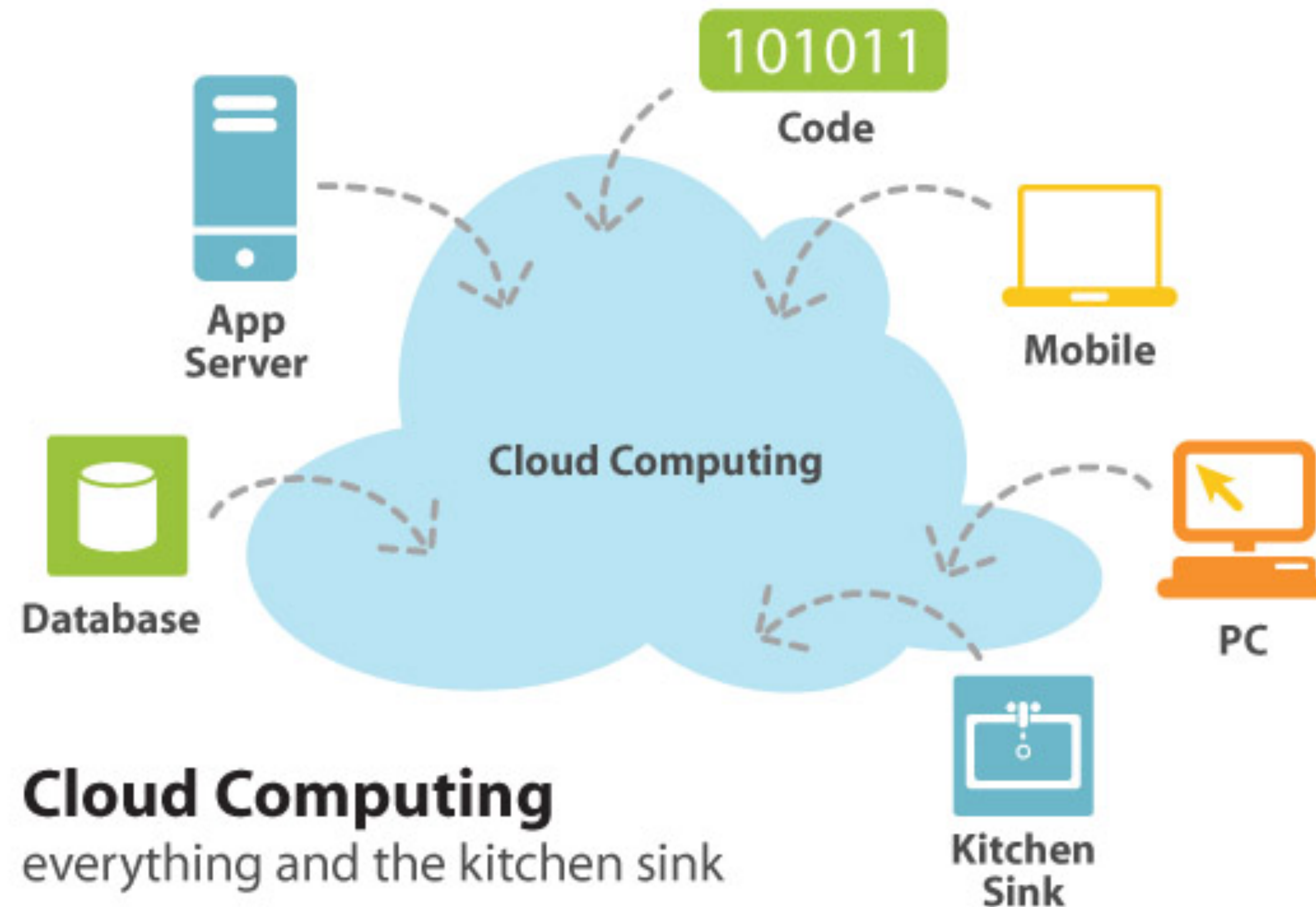
Scale up
and down



Self-service
infrastructure

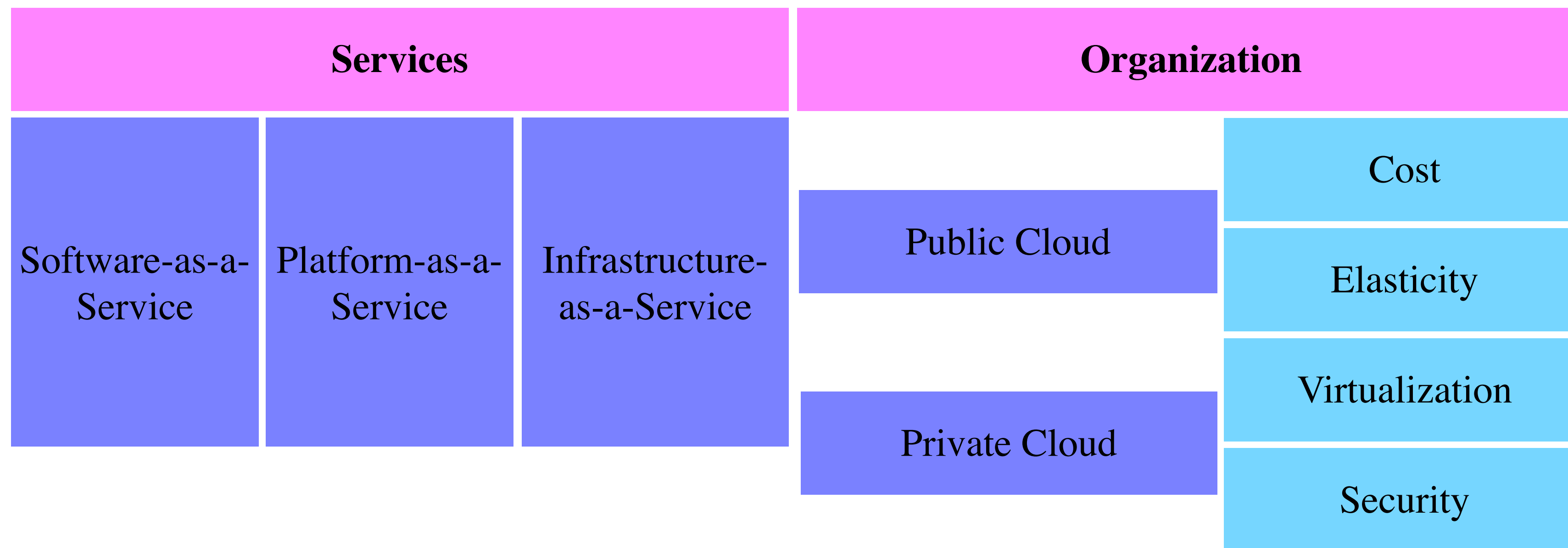
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Cloud computing is the use of computing resources (hardware and software) that are delivered as a service over a network.

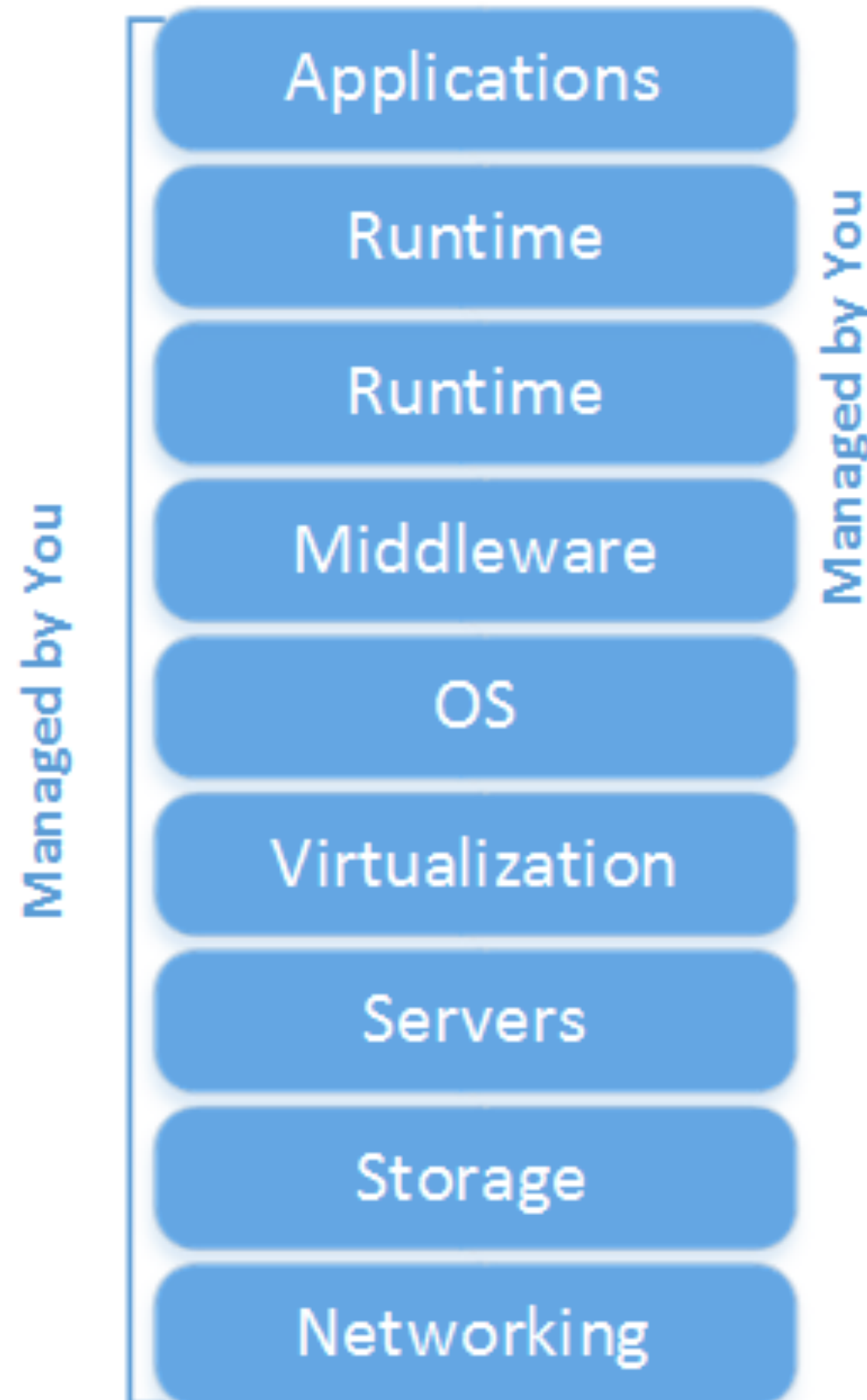


Is Cloud Computing a **really** new technology?

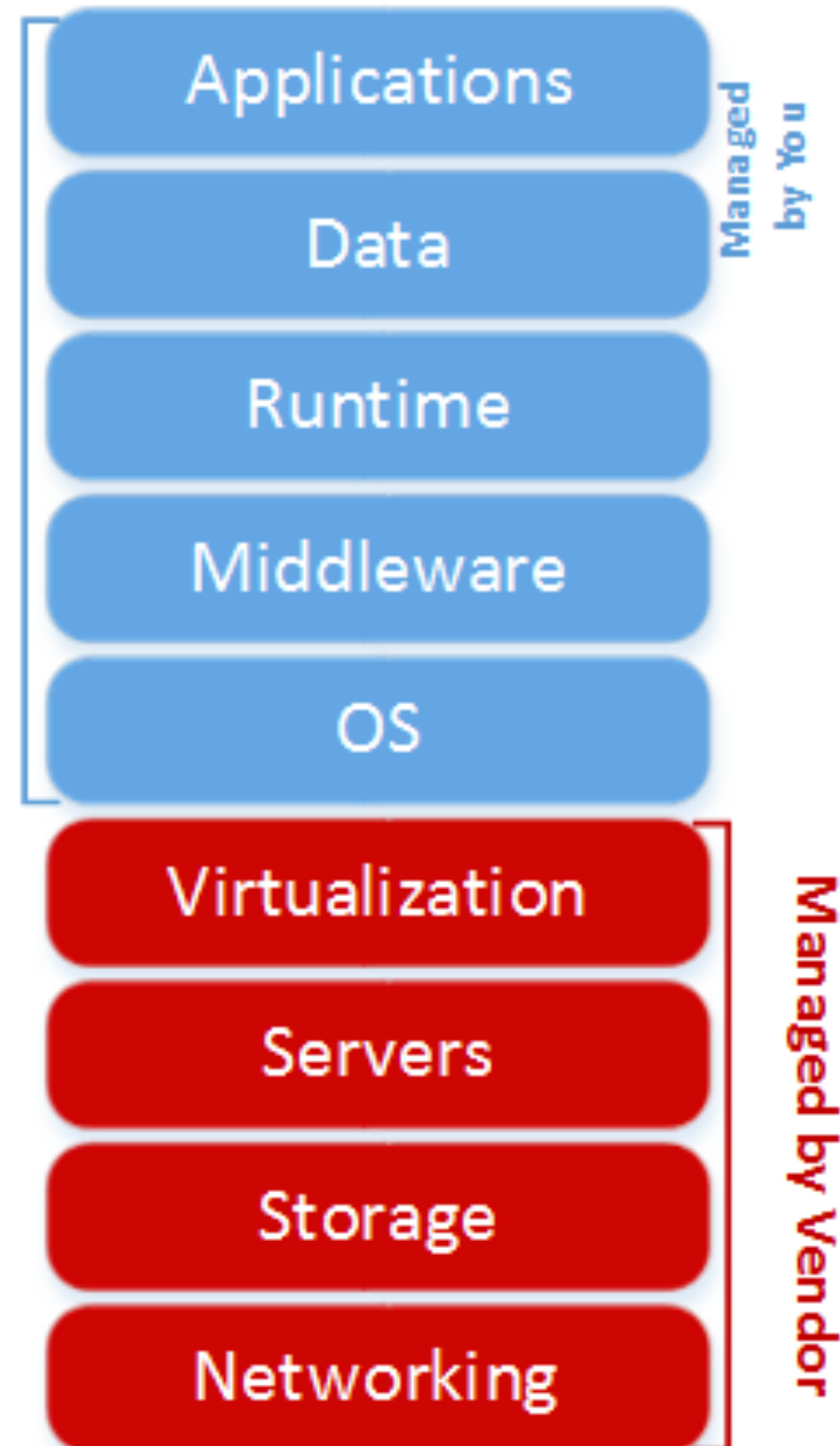
Today, Cloud computing is really linked with IT (Information Technology) decisions. However, We can observe that more and more, its concepts gain space on the life of the most Internet users.



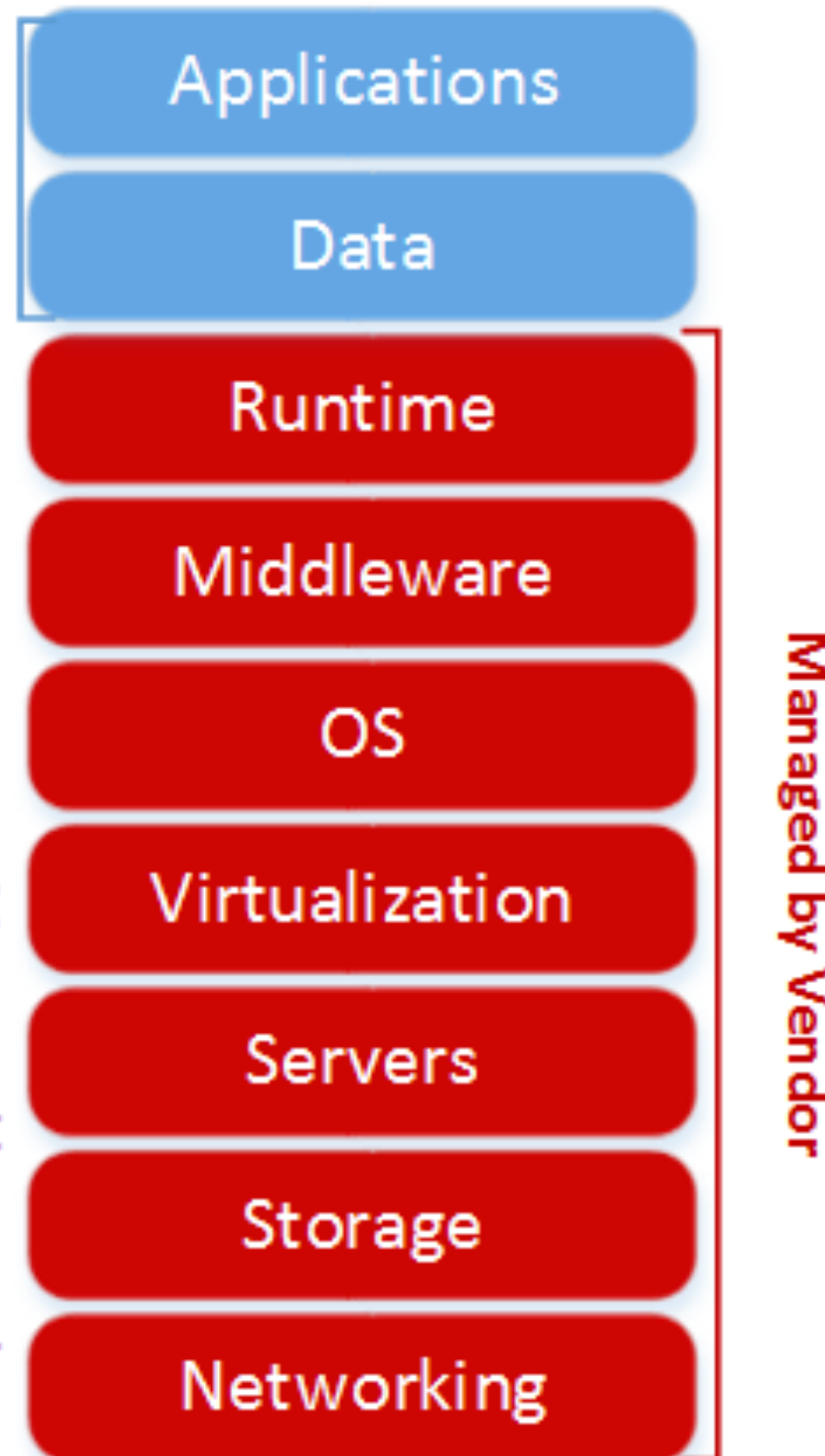
On Premise



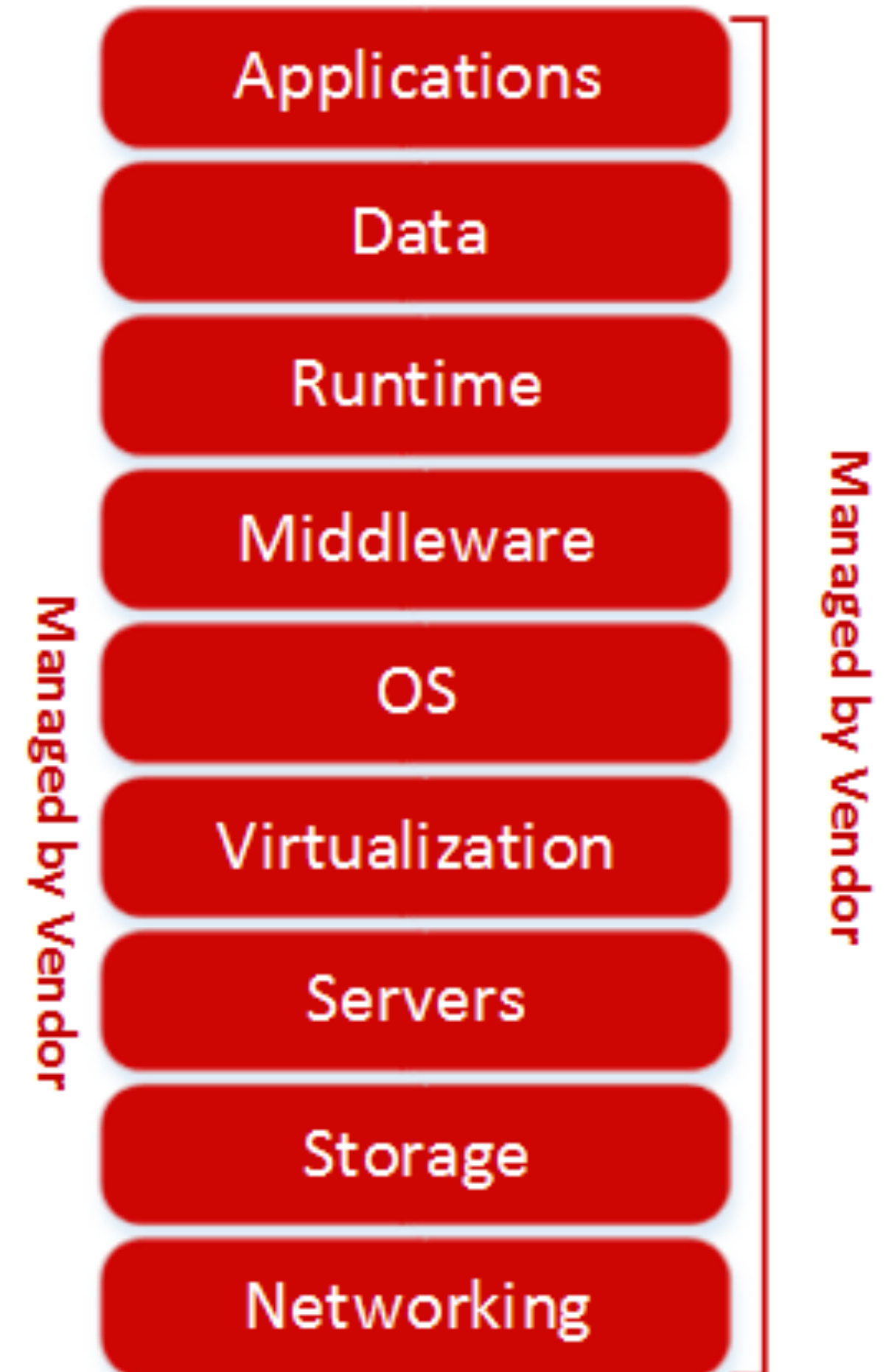
IaaS: Infrastructure as a Service

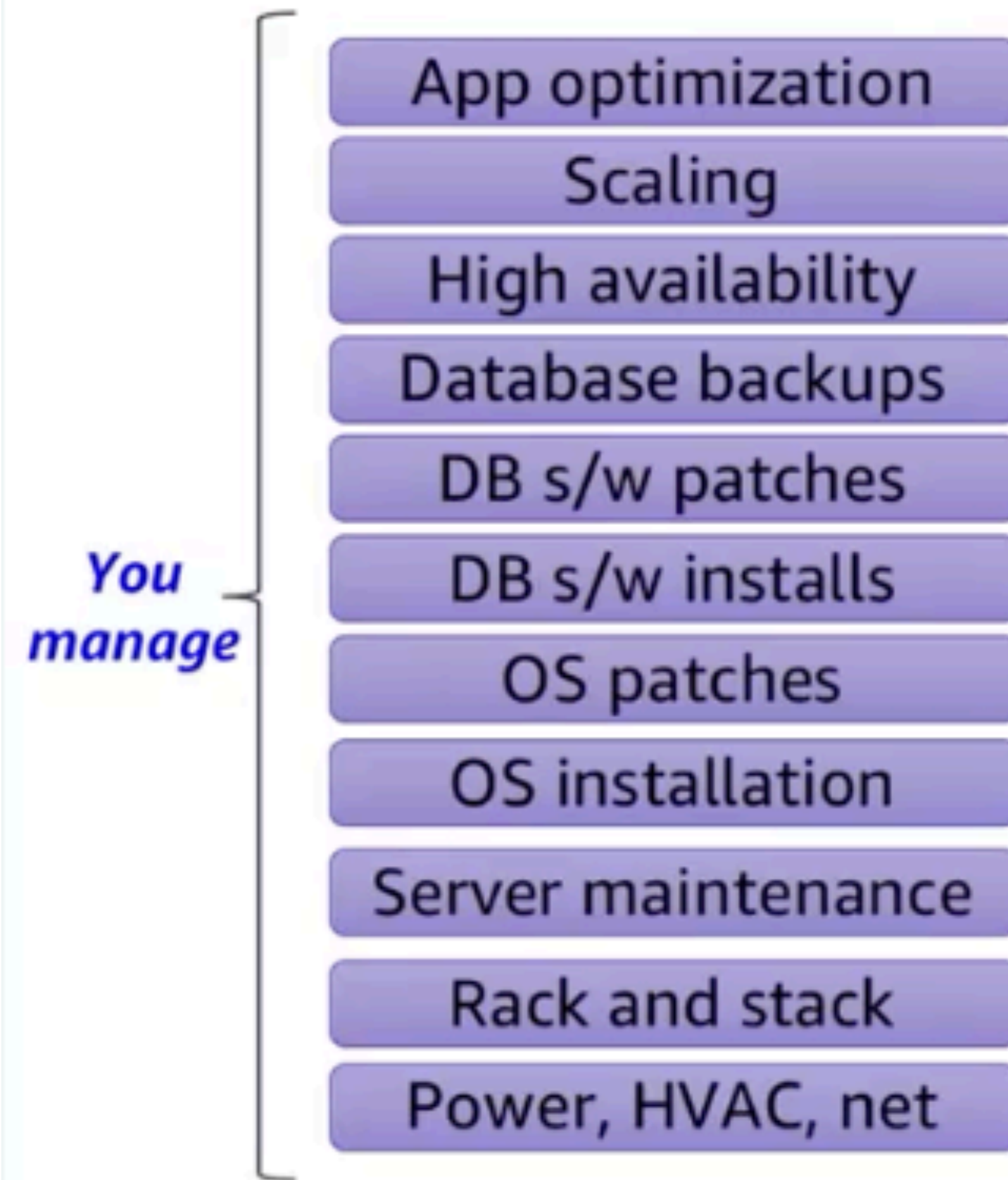


PaaS: Platform as a Service

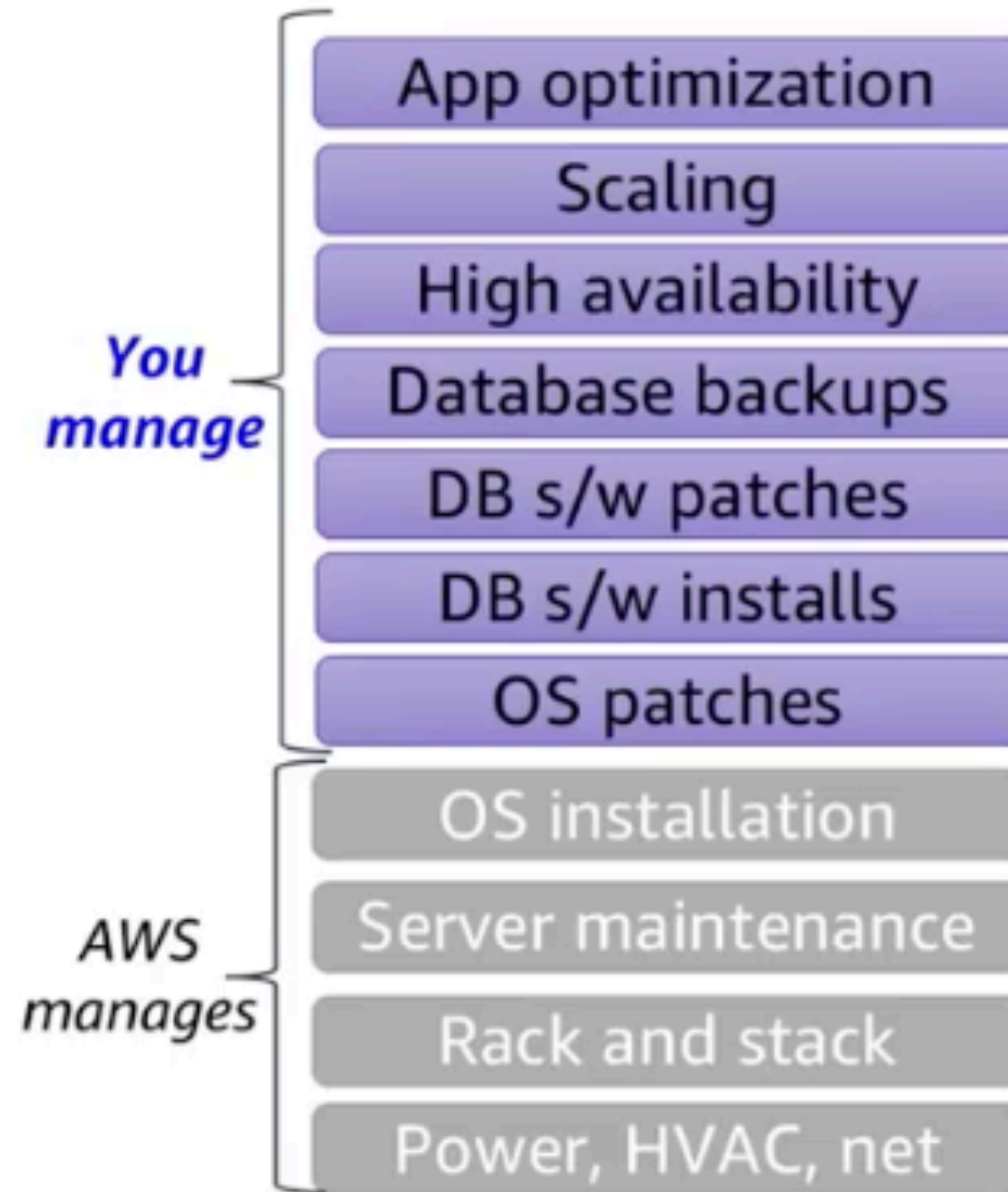


SaaS: Software as a Service

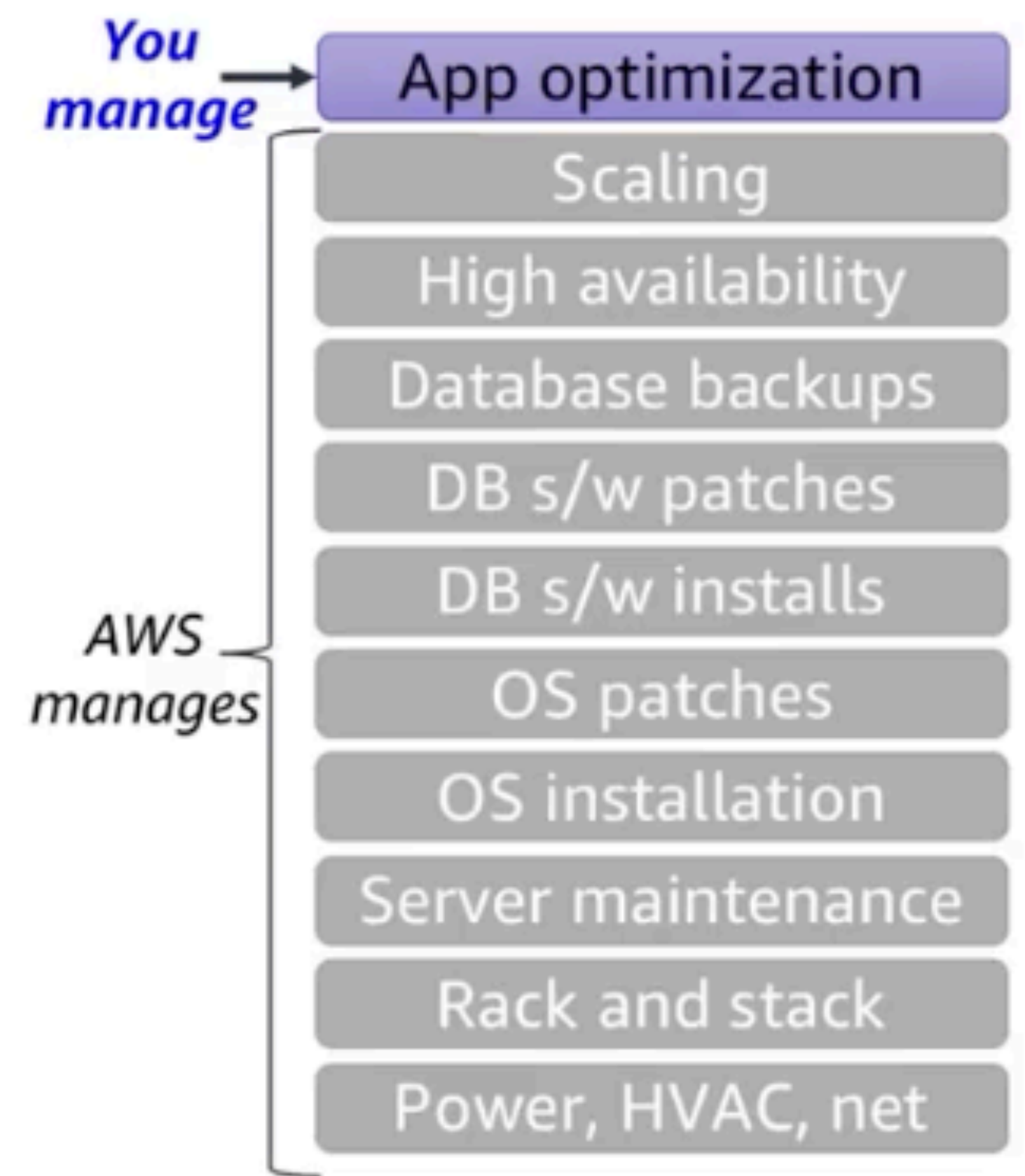




Host database
on-premises



Host database in
Amazon EC2



Host database in
a **managed AWS
database service**

Why We may use cloud computing?

Storage

Pay-as-you-go concept

Fast environment deployment

Virtualization

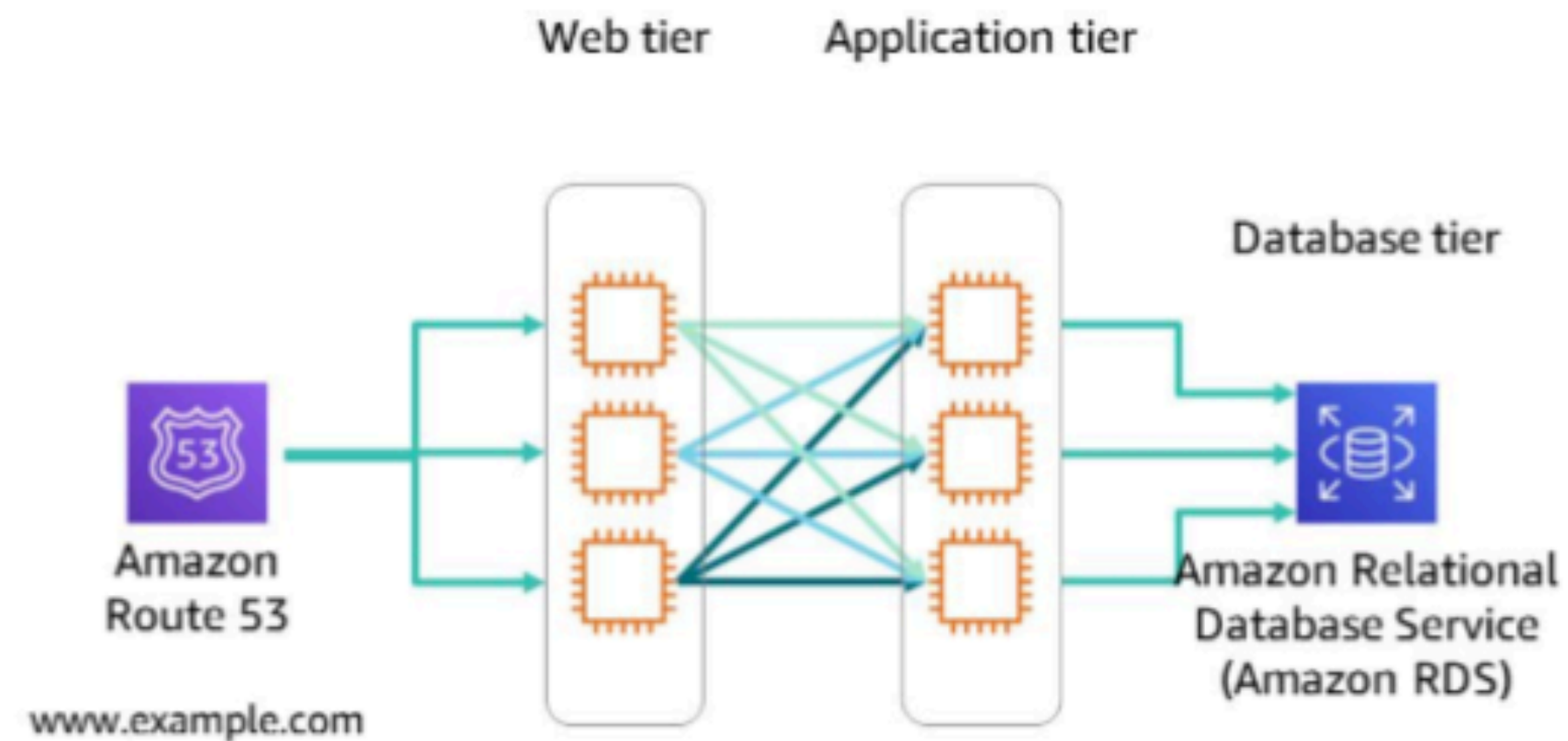
Elasticity

Pros and Cons



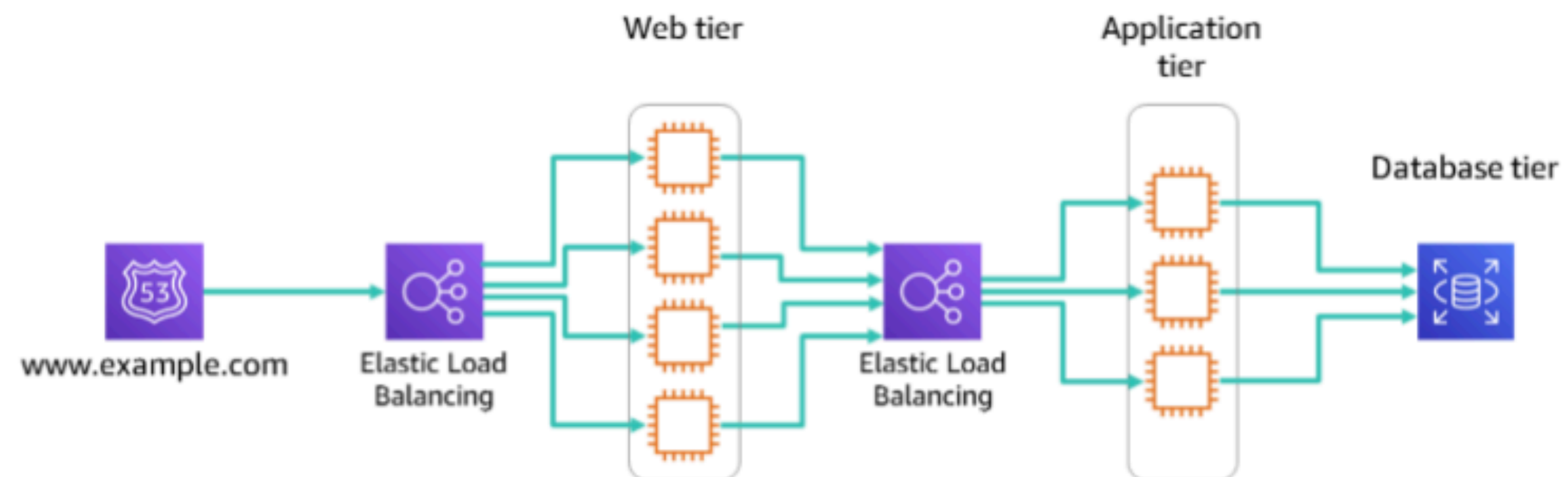
Acoplamentos de Soluções em Nuvem

Tightly coupled architectures



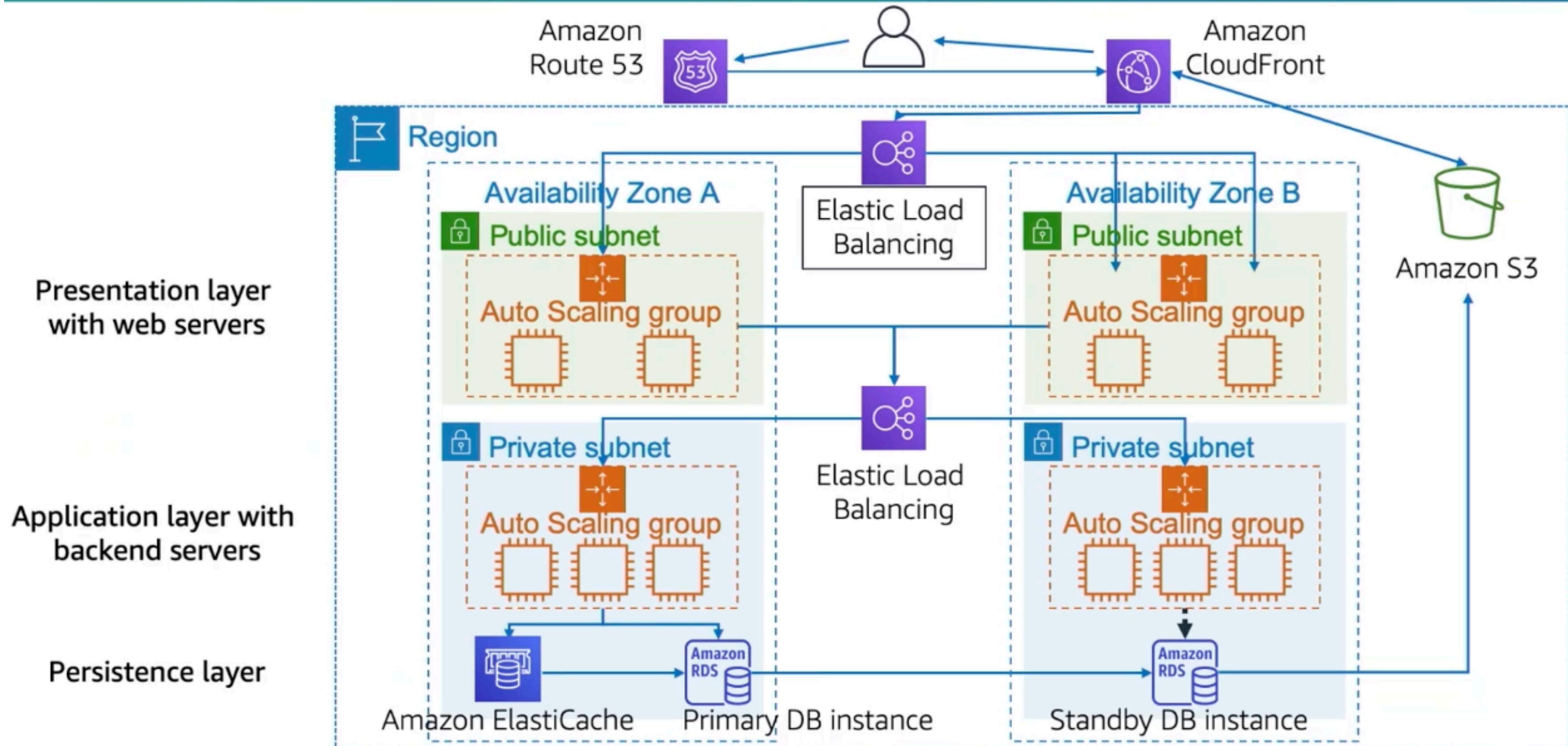
Components are **strongly connected** to each other.

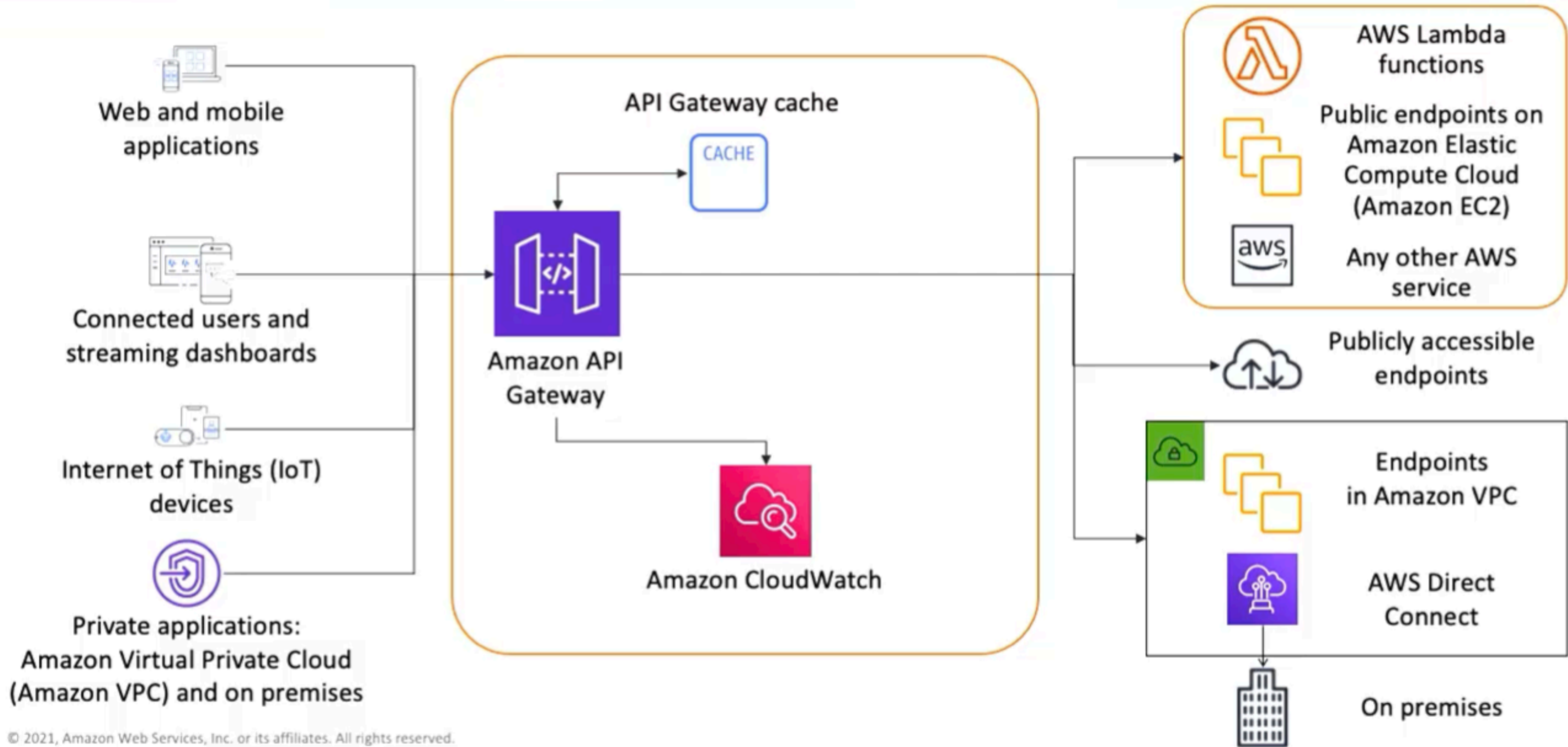
Loosely coupled architectures



Use **managed solutions** as **intermediaries** between layers.

Three-tier web hosting architecture





Feature	Cloud computing	Fog computing
Data processing location	Centralized in remote data centers	Decentralized, at or near the source of data
Latency	Higher due to longer distance data travels	Lower, as data are processed closer to their origin
Bandwidth usage	Higher, as large amounts of data are transmitted to and from the cloud	Lower, due to local data processing and reduced data transmission
Real-time capability	Limited by network latency	Enhanced, suitable for time-sensitive applications such as emergency response and critical care monitoring
Security risks	Higher, due to extensive data transmission and centralized storage	Reduced, with data processed and often stored locally, minimizing exposure to cyber threats
Interoperability	Dependent on internet and cloud services' protocols	Requires local interoperability standards but is less dependent on external networks
Scalability	Easily scalable with cloud resources	Scalability depends on local infrastructure capabilities
Cost	Potentially lower upfront, with ongoing operational expenses	Higher upfront costs for local infrastructure, potentially lower operational costs

Challenges

Safeguarding Data Security

Context-Aware Computing

Managing the Contractual Relationship

Open Standards

Professionals with expertise on Big Data

High Performance Computing

Dynamic applications with large input data

Automatic Elastic Cloud

Talking about cloud elasticity

One of the
most
commented
feature

Actually, It happens not automatically

It depends on user intervention

Writing of
rules and
actions

Text or graphical
interface for proving
resource elasticity

Focus

Load Balancing for Web
Servers

OpenNebula Sunstone

Documentation | Support | Community

Welcome oneadmin | Sign Out

Submitted

VM migrate: 137 >> 15

Virtual Machines

Show	10	entries
☐	All	
☐	134	oneadmin vm01 ACTIVE 0 0K
☐	135	oneadmin vm02 ACTIVE 0 0K
☐	136	oneadmin vm03 ACTIVE 0 0K
☒	137	oneadmin vm05 FAILED 0 0K

Showing 1 to 4 of 4 entries

Virtual Machine Log - vm05

Thu Feb 17 12:30:43 2011 [DiM][I]: New VM state is ACTIVE.
Thu Feb 17 12:30:43 2011 [LCM][I]: New VM state is PROLOG.
Thu Feb 17 12:30:43 2011 [TM][E]: prolog, undefined source disk image in VM template
Thu Feb 17 12:30:44 2011 [DiM][I]: New VM state is FAILED
Thu Feb 17 12:30:44 2011 [TM][W]: Ignored: TRANSFER SUCCESS 137 -

Error

Method: VirtualMachineMigrate
Action: MANAGE
Object: VM
Id: 137
Reason: VM in wrong state

[close all]

OpenNebula Sunstone

Dashboard

oneadmin

Storage

4 IMAGES 4GB USED

Users

2 USERS 2 GROUPS

Network

1 VNETS 4 ADDRESSES

Hosts

1 TOTAL 1 ON 0 OFF 0 ERROR

CPU

MEMORY

Virtual Machines

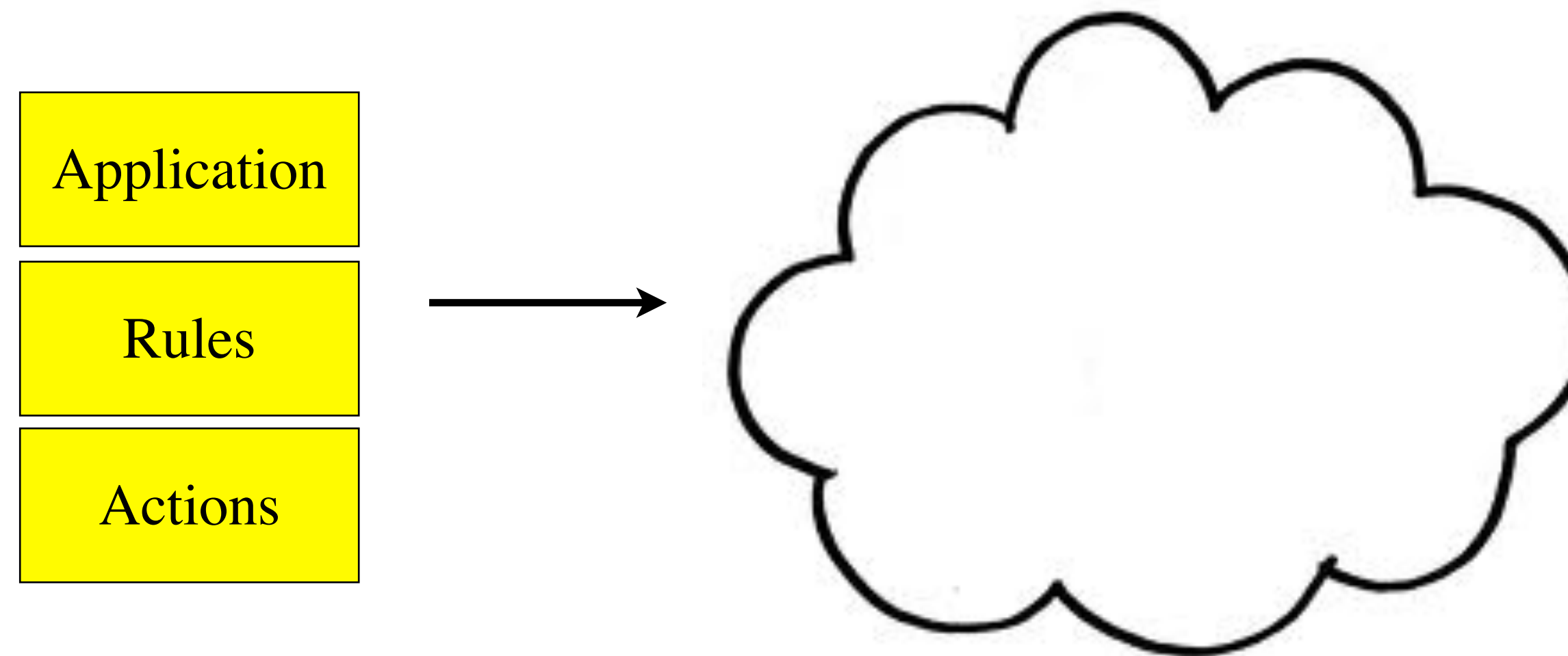
4 TOTAL 4 ACTIVE 0 PENDING 0 FAILED

NET DOWNLOAD SPEED

NET UPLOAD SPEED

Allocated Real Total

Elasticity in the Cloud



Elasticity in the Cloud

```
Template vmtemplate = new Template(templateid, oneClient);
OneResponse rc = vmtemplate.instantiate();
VirtualMachine vm = new VirtualMachine(Integer.parseInt(rc.getMessage()), oneClient);
vm.deploy(hostid);
```

=====

Código do método real que está em produção:

```
public int deploy(Client oc, int hostid, JTextArea plog){
    Template vmtemplate = new Template(this.templateid, oc);
    OneResponse rc = vmtemplate.instantiate();
    log = plog;
    if (!rc.isError()){
        this.id = Integer.parseInt(rc.getMessage());
        this.vm = new VirtualMachine(this.id, oc);
        vm.deploy(hostid);
    }
    return this.id;
}
```

Code to create a new VM

Elasticity in the Cloud

```
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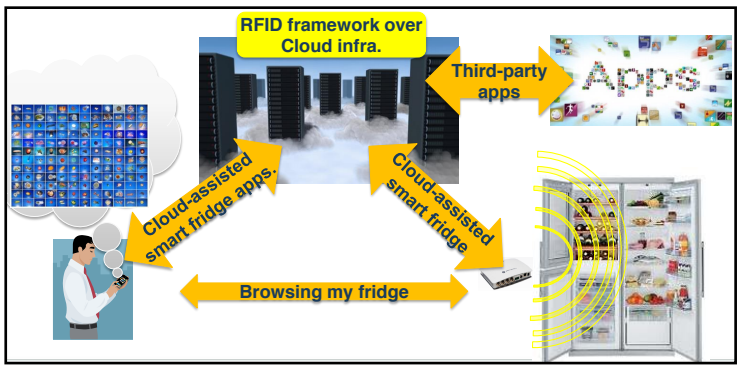
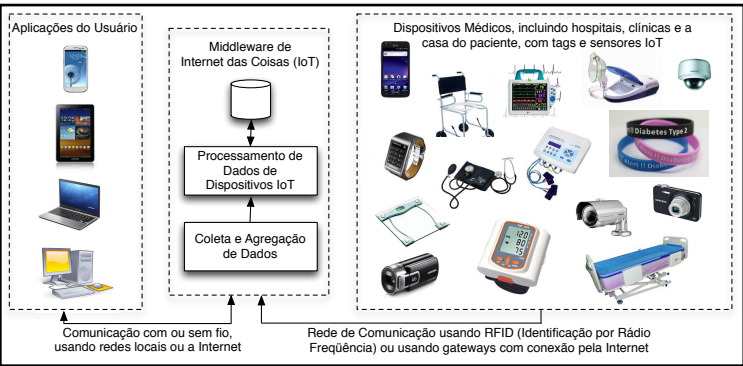
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Code to create a new VM

Research Projects



Trends

Cloud-Centric Design Becomes a Necessity

Opportunistic and Mobile Cloud

Virtual Desktop Infrastructure

Integration of Cloud and Internet of Things

Hybrid Clouds

Cloud Computing Influences Future Data
Centers

Importance of Cloud Brokers will Grow

Green Computing

Extreme low-energy servers

Categorizing compute services

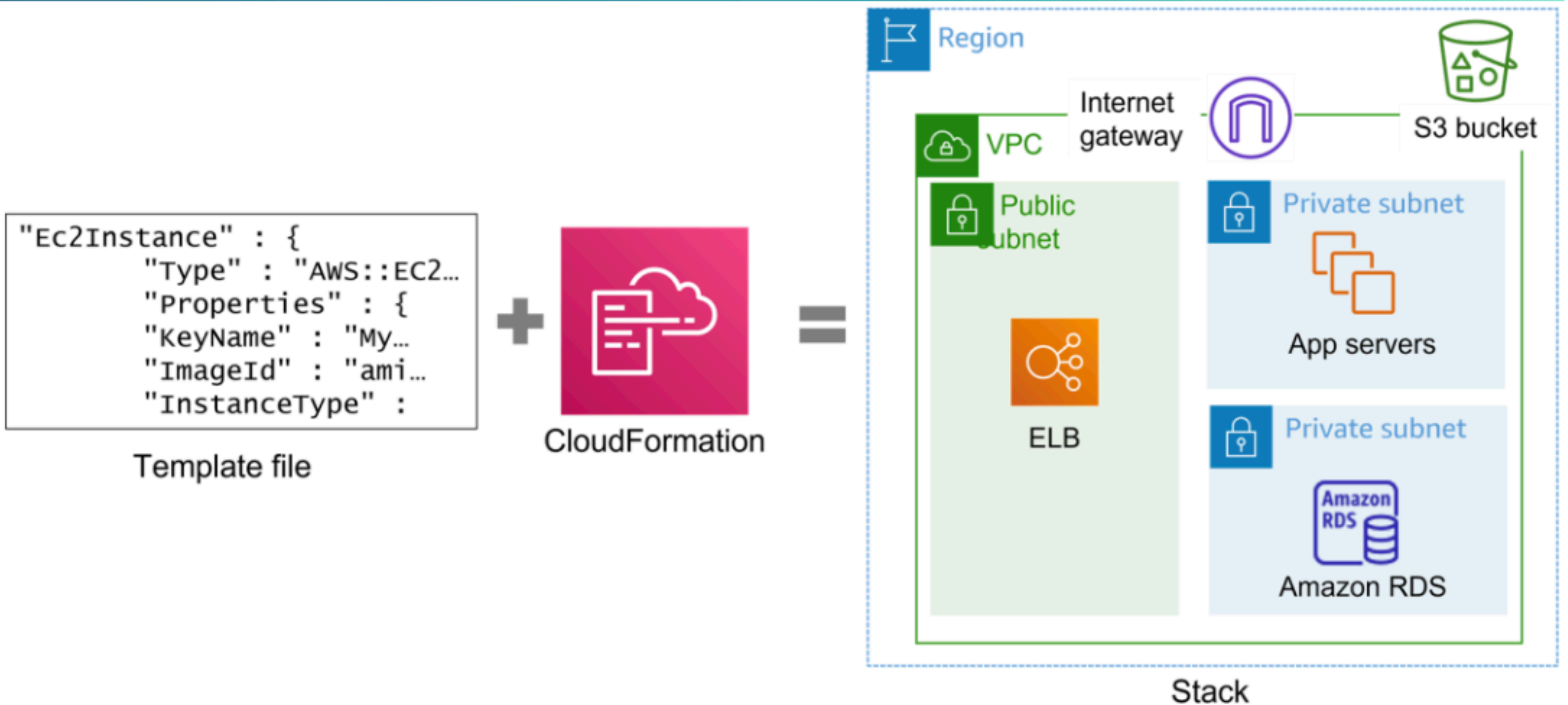
Services	Key Concepts	Characteristics	Ease of Use
Amazon EC2	- Infrastructure as a service (IaaS) - Instance-based - Virtual machines	Provision virtual machines that you can manage as you choose	A familiar concept to many IT professionals.
AWS Lambda	- Serverless computing - Function-based - Low-cost	- Write and deploy code that executes on a schedule or that can be triggered by events - Use when possible (architect for the cloud)	A relatively new concept for many IT staff members, but easy to use after you learn how.
- Amazon ECS - Amazon EKS - AWS Fargate - Amazon ECR	- Container-based computing - Instance-based	Spin up and execute jobs more quickly	AWS Fargate reduces administrative overhead, but you can use options that give you more control.
AWS Elastic Beanstalk	- Platform as a service (PaaS) - For web applications	- Focus on your code (building your application) - Can easily tie into other services—databases, Domain Name System (DNS), etc.	Fast and easy to get started.

Storage comparison



	Amazon S3	Amazon S3 Glacier
Data Volume	No limit	No limit
Average Latency	ms	minutes/hours
Item Size	5 TB maximum	40 TB maximum
Cost/GB per Month	Higher cost	Lower cost
Billed Requests	PUT, COPY, POST, LIST, and GET	UPLOAD and retrieval
Retrieval Pricing	¢ Per request	¢¢ Per request and per GB

How CloudFormation works





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